THE PROBLEM

AI agents can pay. They can’t manage API keys.

Human-shaped billing breaks agents

- Subscriptions and dashboards assume a human operator
- API keys are secrets agents shouldn’t hold long-term
- Per-call on-chain gas kills micropayments

What machine-to-machine needs

- Escrow once, pay incrementally
- Cryptographic vouchers, not passwords
- Provider batch settlement without trusting the agent after deposit

WHY PORTALDOT V3

EVM apps on a Substrate-native stack

Execution

Portaldot V3 exposes an EVM-compatible layer (pallet-revive). Solidity contracts ship with familiar tooling — Foundry, cast, viem — while staying on Portaldot's testnet.

Native tPOT tabs

AgentTab escrows native **tPOTv3** (14 decimals on Substrate) or any ERC-20. Middleware for agent payments fits the Modular Cloud track.

Interop path

Roadmap: settle via `Revive.call` from native `5…` accounts; Assets-pallet tokens as tab currency.

Network facts

Chain ID `420420777`
RPC `test-evm-rpc-v3.portaldot.dev`
Console node-console.portaldot.dev

SOLUTION

AgentTab — escrow + EIP-712 vouchers

Agents escrow tPOT once, sign a voucher per API call; providers batch-settle on Portaldot V3 — micropayments without API keys or per-call gas.

open

Payer deposits native tPOT (or ERC-20) for a provider, with expiry. One tab, one escrow.

voucher

Each call: agent signs EIP-712 `(tabId, cumulative)`. Provider returns HTTP 402 if missing/invalid.

settle

Provider submits cumulative + sig once for N calls. Contract pays `cumulative − claimed`.

refund

After expiry, payer reclaims unspent balance. Reentrancy guard + existence checks on every path.

ARCHITECTURE

Agent → Provider → AgentTab on Portaldot V3

01
AI agent
open(provider, NATIVE, deposit, expiry) with tPOT value

→

02
API provider
request + voucher sig; respond or HTTP 402

→

03
AgentTab.sol
settle batches N calls; refund after expiry

Status (2026-10-22, Asia/Shanghai)

`forge test` → 7/7 passing (ERC-20 + native tabs, bad/stale/exceeds sig, refund, top-up). Ported from author's MIT AgentTab (Celo); Portaldot-specific: native tabs, `call{value:}` payouts, reentrancy guard.

LIVE TESTNET PROOF

Deployed on Portaldot V3 · chain 420420777

Contract	`0x9a096C216f79e08C2DB95d92F412697D8F1f9329`
Deploy tx	`0xe8a4376602a5467574dd7f2e966006da4fe41275d28a989e33f116fa57baab72`
Demo open	`0xc27ab271b49103a257046c7458805ed1c73988d464f61aa98bfe6338a13e534e`
Demo settle	`0xd58678d04863861cda814284eb2879d0abcaf85ca64779cc5e69f2a670b8ac5b`
Deployer	`0x9e2e2024c7dcA86e57608e2ceAD160081c381fDd`
Explorer	node-console.portaldot.dev · see also `DEPLOYMENT.md`

Native tPOT tab #1: deposit 1e16 wei → settle cumulative 5e15 wei via EIP-712 → remaining 5e15.

DEMO FLOW

open → voucher → settle (on-chain today)

OPEN
Tab #1
payer = deployer
provider = 0x7F22...3369
deposit = 1e16 wei native

→

VOUCHER
EIP-712
agent signs (tabId=1, cumulative=5e15)
off-chain, zero gas

→

SETTLE
Provider tx
pays 5e15 to provider
remaining(1) == 5e15
hash 0xd58678d0...ac5b

Reproduce locally

```
forge test && cp .env.example .env && scripts/deploy.sh
```

Full hashes and redeploy notes: `DEPLOYMENT.md` · static replay: `docs/demo-checkin.mp4`

ROADMAP / OUTSTANDING

What’s next for final submission

Provider gateway + SDK

HTTP 402 → EIP-712 voucher gateway. TypeScript agent SDK (viem, chain 420420777, 14-decimal helpers).

Substrate interop

Settle via Revive.call from native 5... accounts. Assets-pallet token as tab currency (ERC-20 precompile or wrapped mock).

Dashboard

Web UI for tabs / vouchers / settlements with node-console deep links.

Done for Week 2

Contract live, open/settle proven, public repo + check-in assets (this deck + ≤3 min demo video).

ASK

Continued support & office hours

We’re asking mentors / Portaldot for:

1) Feedback on native-token gas + `eth_estimateGas` quirks on V3 eth-rpc
2) Guidance on Revive.call / SS58 ↔ EVM address mapping for agent wallets
3) Office hours to pressure-test the HTTP 402 gateway design before final BUIDL polish

Public repo · live contract · demo txs already on-chain.
github.com/zwang1111/agenttab-portaldot · BUIDL 50150

Thank you — AgentTab × Portaldot